TypeMaster: Pro Typing Platform

Real-Time Telemetry, Anti-Cheat Validation & Dual-Engine Supabase Architecture

DEVELOPER
Deepthi Priya

PROJECT STATUS
Production-Ready (v1.0)

CODEBASE VOLUME
40+ Files / 8,500+ LOC

GENERATED DATE
October 08, 2026

01 Executive Summary

TypeMaster is a high-performance, full-stack typing speed test application engineered to deliver professional-grade typing telemetry, interactive gamification, and robust cloud data persistence. Built with modern web technologies (React 19, Vite, Supabase PostgreSQL, Web Audio API), the platform eliminates synthetic or placeholder metrics by implementing a **100% dynamic, real-time data engine** that recalculates typing velocity (WPM), accuracy, streaks, and milestones with sub-millisecond precision.

10	100%	4 Sound	4-Tier
CORE APP VIEWS	LIVE DYNAMIC DATA	AUDIO SWITCH PROFILES	ANTI-CHEAT CHECKS

02 Technical Stack & Architectural Layers

The system is designed with a modular separation of concerns spanning the presentation, real-time computational engine, and cloud database layers:

LAYER	TECHNOLOGY	ROLE & IMPLEMENTATION DETAILS
Frontend Core	`React 19` + `Vite 8`	Ultra-fast HMR, component lifecycle optimization, hooks-based state orchestration across views.
Design System	`Vanilla CSS` (Design Tokens)	Custom design tokens, glassmorphism, responsive CSS grid, zero bloat, custom micro-interactions.
Cloud Database	`Supabase (PostgreSQL)`	Row-Level Security (RLS), tables for users, logs, streaks, achievements, prompts, and auth sessions.
Audio Synthesis	`Web Audio API`	Procedural waveform audio synthesis generating realistic zero-latency mechanical keyboard clicks.
Anti-Cheat Engine	`Custom JS Algorithms`	Sanity checking keystroke intervals, duration validation, anomaly flags, and typing ceiling caps.
Visual Telemetry	`SVG Vector Graphs`	Dynamic, scalable Bezier path interpolation for real-time WPM progression & accuracy trends.

03 Key Features & Systems Built Till Now

⚡ 1. Real-Time Typing Engine

- **Real-time WPM Calculation:** Computed dynamically using standard character formula: `(chars / 5) / (seconds / 60)`.
- **Raw vs. Net WPM:** Accurately isolates gross keystroke speed versus penalized uncorrected errors.
- **Smooth Caret & Live Highlighting:** Character-by-character feedback (green for correct, red for incorrect, pulsing active cursor).
- **Configurable Test Modes:** Durations of 1m, 3m, 5m, and Custom; Easy, Medium, and Hard difficulties.

🎹 2. Acoustic & Virtual Keyboard

- **Procedural Sound Synthesizer:** Zero-dependency Web Audio API implementation with 4 tactile switch modes (Cherry Blue, Cherry Red, Brown, Typewriter).
- **Interactive Virtual Keyboard:** Real-time keypress highlighting on visual QWERTY keyboard with home-row finger indicators.
- **Audio Feedback:** Auditory error buzzer on mistakes and 4-note ascending celebratory chime upon test completion.

🛡️ 3. Anti-Cheat & Verification Service

- **Duration Threshold Check:** Rejects tests terminated in < 3 seconds to prevent stat dilution.
- **Human Feasibility Limit:** Flags and rejects any anomaly exceeding human physiological limits (> 250 WPM).
- **Character Ratio Sanity:** Validates reported character counts against time elapsed.
- **Automated Grading:** Awards performance badges (Needs Practice, Good, Great, Excellent).

📊 4. Telemetry & Analytics Dashboard

- **5 Real-Time Stat Cards:** Best WPM, Average WPM, Overall Accuracy, Tests Completed, Day Streak.
- **Dynamic SVG Trend Charts:** Plots historical test sessions without third-party chart dependencies.
- **Streak Engine:** Automatically calculates consecutive active practice days and longest streaks.
- **Test Logs History:** Real-time session ledger with detailed drill-down modal.

04 Database Schema & Cloud Persistence

A complete production-grade relational database schema was authored in `supabase_schema.sql`, featuring full Row Level Security (RLS) policies and dual-mode runtime compatibility:

TABLE NAME	PRIMARY KEY	CORE COLUMNS / METRICS TRACKED	SECURITY & POLICIES
`users`	`id (TEXT)`	username, email, best_wpm, avg_wpm, accuracy, tests_completed, is_premium	RLS enabled; authenticated read/write
`test_logs`	`id (UUID)`	user_id, wpm, raw_wpm, accuracy, duration, errors, category, difficulty, badge	RLS enabled; cascade delete on user removal
`streaks`	`id (UUID)`	user_id, current_streak, longest_streak, last_active_date, updated_at	RLS enabled; daily timestamp tracking
`achievements`	`(id, user_id)`	title, description, icon, unlocked (boolean), unlocked_at	RLS enabled; dynamic trigger checks
`prompts`	`id (UUID)`	category, difficulty, text, word_count, created_at	Public read; categorized seed library

Dual-Engine Architecture: The application incorporates an embedded local engine fallback. If Supabase cloud credentials are not yet entered, TypeMaster transparently persists all live metrics to browser storage and allows immediate full testing without blocking the user. When Supabase keys are connected via the in-app modal, the app seamlessly integrates with the remote PostgreSQL instance.

05 Complete View & Navigation Ecosystem

The platform features 10 fully realized application views, structured in an intuitive sidebar navigation layout:

VIEW COMPONENT	FUNCTIONAL HIGHLIGHTS	STATUS
DashboardView.jsx	Real-time user welcome, 5 live metric cards, dynamic SVG telemetry charts, quick test launcher, and recent sessions table.	Completed & Real-Time
TypingTestView.jsx	Immersive test arena, word stream, smooth caret, live WPM counter, audio feedback, virtual keyboard, and completion confetti.	Completed & Real-Time
PerformanceView.jsx	In-depth telemetry analytics, speed distribution tiers, accuracy breakdown, and consistency score calculations.	Completed & Real-Time
ProgressView.jsx	Gamification hub, active day streak counter, milestone achievement cards, dynamic XP bar, and level status.	Completed & Real-Time
LeaderboardView.jsx	Global community typist rankings, filterable by time horizon and difficulty, highlighting personal percentile rank.	Completed & Real-Time
PracticeView.jsx	Targeted drills, tricky key combinations, number row focus, and programming syntax training mode.	Completed & Real-Time
HistoryView.jsx	Searchable, filterable audit log of all completed tests with instant session detail inspection modal.	Completed & Real-Time
SettingsView.jsx	Theme customization (Cyberpunk, Matrix, Dracula, Slate), switch sound selector, volume slider, and Supabase config.	Completed & Real-Time
LoginView.jsx	Dedicated Supabase email/password login, account registration, error handling, and instant Guest mode access.	Completed & Real-Time
HelpView.jsx	Touch typing guide, finger placement ergonomics, home row tips, and platform documentation.	Completed & Real-Time

06 Gamification & Milestone Unlocks

Milestones are dynamically evaluated at the conclusion of every validated test submission:

🏆 Speed Master (75+ WPM)

Unlocked automatically once the user's verified net typing speed crosses 75 words per minute.

🎯 Accuracy King (98%+ Accuracy)

Awarded when a user completes a typing test with an exceptional accuracy threshold of 98% or higher.

🔥 10-Day Streak (Daily Practice)

Triggers when a typist maintains 10 consecutive days of validated practice sessions.

⚡ Lightning Fast (100+ WPM)

Elite tier badge granted upon breaking the triple-digit speed milestone in any official test.

07 Repository Audit & Commit Milestones

The project repository is structured cleanly with semantic git commits documenting every progressive phase:

COMMIT	COMMIT MESSAGE / SCOPE	KEY CHANGES
`7212459`	**feat: scaffold full-featured typing master web application**	Scaffolded Vite + React project; implemented all 10 views, VirtualKeyboard, Web Audio sound synthesizer, AntiCheat service, and Supabase SQL schema (41 files, +8,587 lines).
`ede3656`	**docs: update README with real-time features and remove test script**	Cleaned development test scripts; updated documentation with Supabase setup guide and feature lists.
`aea19ca`	**feat: make performance, progress, and streaks dynamic and real-time**	Removed static mock data fallbacks; wired pure real-time calculations for Best WPM, Average WPM, Accuracy %, Streaks, and dynamic SVG graph plotting across views.

08 Future Roadmap & Upcoming Capabilities

🏎️ Real-Time Multiplayer Racing

Leveraging Supabase Realtime Channels to facilitate live head-to-head typing competitions with race avatars and instantaneous progress indicators.

📜 Verified Typing Certificate

One-click generation of verifiable, cryptographic typing speed certificates with QR validation and anti-cheat checksums.

🤖 AI Weakness Diagnostic

Heuristic clustering algorithm analyzing specific character pairs (bigrams/trigrams) that trigger user deceleration, generating custom drill passages.

📱 Mobile Touch & Tablet Optimization

Custom software on-screen split keyboards and adaptive layout optimization for touch-based mobile and tablet typing tests.

Conclusion: The TypeMaster project has reached full production-ready capability. All core pillars—from the real-time typing engine, procedural audio synthesizer, anti-cheat validation, and Supabase cloud synchronization, to the 10 dedicated views and responsive design system—are complete, verified, and running smoothly.